

Agile Software Development

Core Concepts Summary Notes • 11 May 2026

1. Core Agile Concepts

Agile	A philosophy that breaks projects into small iterations, delivers value continuously, and adapts based on feedback.
Agile Manifesto	The 2001 founding document with four core values and twelve principles: individuals, working solutions, customer collaboration, and adaptability.
Agile Mindset	Collaborative attitudes that treat failures as learning opportunities and embrace change to succeed in dynamic environments.
Acceptance Criteria	Specific conditions a work item must meet for the Product Owner (PO) or customer to accept it as complete.

2. Planning and Execution

Backlog	A prioritised list of work items such as features, bug fixes, and user stories awaiting completion by the team.
Epic	A large, high-level user story too big for one iteration; broken down into smaller stories.
Iteration / Sprint	A short, fixed timebox of 1 to 4 weeks where the team completes a set of prioritised work items. Sprint is the Scrum term for iteration.
Minimum Viable Product (MVP)	The simplest product version with just enough features to be usable and gather validated learning from customers.
Story Points	An abstract unit teams use to estimate the relative effort, complexity, and risk of a user story.
Estimation	The process of assigning a value such as story points or hours to a work item to gauge its size and effort.
Refinement (Grooming)	An ongoing activity where the Product Owner (PO) and team review, clarify, and estimate backlog items to prepare for upcoming sprints.
Velocity	A metric tracking total story points completed in a sprint, used for forecasting future capacity.

3. Events and Monitoring

Ceremonies	Key Scrum events including planning, daily stand-up, review, and retrospective that provide structure for the team.
Daily Scrum (Stand-up)	A short daily meeting for the team to synchronise activities and plan the next 24 hours.
Sprint Planning	A collaborative meeting at the start of a sprint to define what can be delivered and how it will be achieved.
Sprint Review	A meeting at the end of a sprint to inspect the increment and adapt the Product Backlog.
Sprint Retrospective	An internal team meeting after the Sprint Review to identify improvement opportunities for the next sprint.
Burndown Chart	A visual chart showing remaining work versus time to track progress within a sprint.

Burnup Chart	A chart showing completed work versus total scope, useful for tracking progress and identifying scope creep.
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4. Frameworks and Methods

Scrum	An Agile framework using fixed-length sprints and specific roles to build and deliver products incrementally.
Kanban	A method that visualises workflow on a board and limits Work In Progress (WIP) to improve flow and efficiency.
Lean	A method focused on maximising customer value while eliminating waste, meaning anything not delivering value.
Extreme Programming (XP)	An Agile methodology emphasising test-driven development, pair programming, and frequent releases to improve quality.

5. Key Roles

Product Owner (PO)	Responsible for maximising product value and managing the Product Backlog by prioritising work.
Scrum Master (SM)	A servant-leader who champions Agile processes, removes impediments, and ensures the team practises Agile effectively.
Agile Team	A cross-functional, self-organising group of typically 5 to 11 individuals who work together to deliver value.

6. Scaling and Organisational Terms

Scaled Agile Framework (SAFe)	A knowledge base for implementing Lean-Agile practices at enterprise scale across large organisations.
Scrum of Scrums	A technique to coordinate multiple Scrum teams by sending representatives to a shared meeting to track cross-team dependencies.
DevOps (Development and Operations)	A practice bridging development and IT operations to shorten the development lifecycle and provide continuous delivery with high quality.
Agile Transformation	The process of gradually aligning an entire organisation's culture, processes, and practices with Agile principles.

7. Common Metrics and Constraints

Bottleneck	A constraint point in the workflow where tasks pile up and slow down the entire process.
Cadence	A consistent, predictable rhythm of events such as sprints starting every two weeks, crucial for planning and feedback.
Quality Triangle	A model showing the constraints of Time, Cost, and Quality in project management, where a change in one impacts the others.
Waste	Any activity that consumes resources without creating value for the customer; a central concept in Lean methodology.
Waterfall	A traditional sequential development model and the conceptual opposite of Agile, often used as a point of comparison.